

Financial Feasibility and Operational Value Realization of Cloud-Native ERP Implementation: ROI Analysis and Multi-Dimension Business Benefit Measurement in Telecom Infrastructure Services

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Abstract—This paper evaluates the financial and operational returns of cloud-native enterprise resource planning system adoption in a large-scale telecommunications service organization. The research employs a multi-year longitudinal financial analysis examining the investment feasibility and business value realization of cloud-based ERP implementation involving substantial capital investment representing a significant percentage of organizational capex budget. Using a mixed-methods approach combining comparative financial statement analysis, quantitative benefit categorization through Ranti's Generic IT/IS Business Value framework, and rigorous Cost-Benefit Analysis methodology, the study measures investment returns across multiple dimensions. Financial metrics reveal positive investment viability: Net Present Value exceeding zero, Return on Investment percentage on total asset base, capital payback period of less than one year, and Benefit-Cost Ratio exceeding breakeven threshold. Comparative analysis of consolidated financial statements demonstrates significant operational improvement: profit growth acceleration percentage in year-end results, sustained revenue growth trajectory, proportional expansion in total asset base, and enhanced equity position. Benefit categorization identifies five critical value realization pathways: accelerated payment processing and liability management, enhanced strategic decision-making capability through unified financial data infrastructure, revenue growth driven by customer confidence expansion, operational cost reduction in document management through digital transformation, and improved organizational reputation through award recognition for data governance excellence. The analysis concludes that cloud-native ERP platforms deliver measurable return on investment within sub-annual timeframes while simultaneously enabling intangible strategic capabilities. Study limitations acknowledge reliance on secondary financial data and recommend future research incorporating primary data collection through management interviews and employee surveys to isolate specific functionality-driven benefits. This research provides evidence-based financial validation supporting executive decision-making for ERP modernization investments in service-intensive industries.

I. INTRODUCTION

A. Digital Transformation Context in Indonesian Telecommunications

The telecommunications industry in Indonesia has experienced fundamental structural transformation over the past decade. Market liberalization, rapid mobile penetration, and

evolving consumer expectations have compelled incumbent operators to re-evaluate their operational infrastructure. PT XYZ, established in 1996 as a private telecommunications service provider, has grown to serve 57.9 million prepaid and postpaid subscribers across the Indonesian archipelago. This scale of operations requires sophisticated enterprise systems capable of managing complex billing, customer relationship management, network asset accounting, and regulatory compliance obligations.

The competitive landscape intensified following the entry of new market participants and the convergence of telecommunications, media, and technology sectors. Traditional circuit-switched voice revenue declined as over-the-top messaging applications gained widespread adoption. Data traffic volumes increased exponentially while average revenue per user experienced sustained compression. These structural pressures created an imperative for operational efficiency gains achievable only through fundamental information system modernization.

B. Strategic Investment Rationale

In 2020, PT XYZ committed substantial capital resources to implement SAP S/4HANA Cloud, representing 20 percent of the company's annual capital expenditure budget or approximately 1.5 trillion Indonesian rupiah. This investment positioned PT XYZ as the first telecommunications operator in Southeast Asia to deploy SAP S/4HANA Cloud as its core enterprise resource planning platform. The decision reflected strategic recognition that legacy systems constrained organizational agility, created data fragmentation across business functions, and limited management's ability to access real-time operational intelligence.

Cloud-native ERP architecture differs fundamentally from traditional on-premise implementations. The S/4HANA Cloud platform operates on SAP's HANA in-memory database, eliminating data redundancy, accelerating transaction processing, and enabling real-time analytics without batch processing delays. The platform's multi-tenant cloud delivery model shifts capital expenditure to operational expenditure, though PT XYZ elected to capitalize the investment as an intangible

asset consistent with Indonesian accounting standards. This accounting treatment enables systematic amortization while maintaining the investment’s visibility in capital allocation decisions.

C. Investment Evaluation Imperative

Enterprise resource planning implementations represent among the most complex and costly information technology investments organizations undertake. Research consistently demonstrates that a substantial proportion of ERP projects experience cost overruns, timeline delays, or failure to deliver anticipated business benefits. The magnitude of PT XYZ’s investment, combined with the strategic visibility of becoming a regional first-mover, created explicit accountability requirements for post-implementation benefit realization assessment.

Traditional information technology investment evaluation methodologies developed for departmental systems prove inadequate for enterprise-scale transformations. Net present value calculations alone cannot capture process improvements, decision quality enhancements, or strategic option value created through integrated information infrastructure. Mixed-method approaches combining quantitative financial metrics with structured benefit categorization provide more comprehensive investment performance assessment.

TABLE I: PT XYZ Operational Scale Pre-Implementation (2019)

Operational Metric	Value
Total Subscribers	57.9 million
Prepaid Subscribers	52.4 million
Postpaid Subscribers	5.5 million
Network Coverage	34 provinces
Cellular Towers	52,000+
Annual Revenue	Rp 25.13 trillion
Employees	4,200+

D. Research Objectives and Scope

This study addresses three primary research objectives. First, to quantify the financial feasibility of cloud-native ERP investment through discounted cash flow methodologies including net present value, internal rate of return, payback period, and benefit-cost ratio. Second, to identify and categorize operational business benefits realized through the implementation using Ranti’s Generic IT/IS Business Value framework, which provides structured classification of information technology benefits across thirteen categories and seventy-four sub-categories. Third, to conduct longitudinal comparative analysis of PT XYZ’s financial statements from 2018 through 2021 to identify performance trends coincident with system implementation.

The research scope encompasses financial data from annual reports, investment cost documentation, and operational benefit evidence collected through archival analysis. The analysis period spans four fiscal years, providing two years of pre-implementation baseline and two years of post-implementation

observation. This timeframe enables identification of performance changes while acknowledging that full benefit realization for enterprise systems typically extends three to five years post-implementation.

E. Research Contributions

This study contributes to the information technology investment evaluation literature in several dimensions. The research provides empirical evidence from the telecommunications sector, which exhibits distinctive characteristics including high fixed asset intensity, real-time operational requirements, and complex regulatory compliance obligations. The analysis demonstrates practical application of Ranti’s framework, originally developed in the Indonesian context, to a large-scale cloud ERP implementation. Financial feasibility calculations incorporate Indonesian macroeconomic parameters including central bank discount rates and local inflation expectations, providing regionally relevant investment benchmarks.

II. RELATED WORK

A. Parallel Developments in AI-Enabled Financial and Operational Systems

Contemporary research in artificial intelligence-enabled financial systems demonstrates parallel methodological concerns with the information technology investment evaluation literature. Mitra developed AI-driven frameworks integrating genetic algorithms with sentiment analysis for mental health care applications, demonstrating that multimodal data fusion techniques effectively capture complex patterns that resist reduction to single-metric assessment [1], [2]. The methodological principle that comprehensive evaluation requires integration of heterogeneous data sources parallels this study’s mixed-methods approach combining financial, operational, and reputational benefit assessment.

Bachu applied time-series analysis and machine learning techniques to energy demand forecasting, demonstrating that predictive accuracy improves substantially when historical patterns are augmented with contextual information [3]. This finding parallels the observation that historical volatility alone provides insufficient basis for investment performance prediction, necessitating incorporation of operational benefit evidence and strategic capability assessment.

Manne developed AI-enhanced payment systems incorporating ARM Trust Zone security architecture and divisible e-cash protocols, demonstrating that enterprise system implementations require concurrent investment in security infrastructure, process redesign, and organizational capability development [4]–[6]. The multi-dimensional benefit realization observed in PT XYZ’s implementation—spanning operational efficiency, decision quality, customer confidence, and reputational enhancement—aligns with Manne’s finding that enterprise system value emerges through multiple complementary mechanisms rather than single-channel cost reduction.

Singh examined real-time IoT data processing architectures and feedback control loops in sensor networks, demonstrating that latency reduction and data integration capabilities generate substantial operational efficiency improvements [7], [8]. The payment processing cycle reduction from fourteen days to three days observed at PT XYZ exemplifies the operational benefits achievable through integrated real-time data architectures documented in Singh’s work.

B. Security, Anomaly Detection, and Process Verification

Pawar investigated secure distributed computing protocols for healthcare information exchange, demonstrating that enterprise system implementations must incorporate security architecture as integral design element rather than post-implementation addition [9]. PT XYZ’s achievement of Indonesian Big Data Association’s Best Data Governance Enterprise Data Protection Excellence award validates the integration of security and data governance capabilities within the SAP S/4HANA Cloud implementation.

Parashar developed real-time anomaly detection systems for financial transaction processing using behavioral pattern analysis and streaming temporal logic verification [10]–[12]. These methodologies for continuous verification of transaction integrity parallel the embedded analytics and real-time monitoring capabilities enabled by SAP S/4HANA Cloud’s in-memory database architecture. The twenty-four-hour financial reporting closure achieved by PT XYZ exemplifies the operational benefits of real-time data processing capabilities.

Banik developed categorical semantics for higher-order process calculi and fixed-point constructions in 2-categories of endofunctors, providing theoretical foundations for concurrent process verification that inform enterprise system reliability assurance [13], [14]. Banik also applied transformer-based curriculum learning approaches to instructional video summarization, demonstrating that modern machine learning techniques effectively extract actionable insights from unstructured textual data [15]. This methodological approach to extracting structured information from unstructured sources parallels the benefit identification methodology employed in this study’s analysis of management discussion and narrative disclosures.

C. Computational Efficiency and Resource Optimization

Koilada investigated hardware-aware neural architecture search, model compression techniques, and efficient pruning methodologies for deep learning inference in resource-constrained environments [16]–[18]. These computational efficiency innovations address challenges parallel to those motivating cloud-native ERP adoption: legacy system performance limitations, resource utilization optimization, and total cost of ownership reduction. The architectural efficiency gains documented in Koilada’s work—reduced computational overhead, improved inference speed, and enhanced resource utilization—parallel the operational efficiency improvements achieved through SAP S/4HANA Cloud’s in-memory database and streamlined data model.

Razdan developed complementary priors and margin-based learning approaches for robust sentiment classification, demonstrating that hybrid AI architectures combining multiple methodological approaches achieve superior performance to single-technique implementations [19], [20]. This architectural principle—that integrated systems outperform point solutions—directly parallels the finding that ERP implementations combining financial, operational, and analytical capabilities deliver greater value than standalone departmental systems.

Abhishek developed cloud-edge digital twin architectures for adaptive battery health management in sustainable transport systems, demonstrating that distributed computing architectures combining centralized cloud analytics with edge processing capabilities achieve optimal performance characteristics [21]–[23]. The cloud-native architecture of SAP S/4HANA Cloud embodies similar design principles: centralized data governance with distributed access, real-time analytics with historical trend analysis, and standardized processes with local configuration flexibility. Abhishek’s transformer-based inference methodology for extracting residential electric vehicle charging patterns from smart meter time series demonstrates that modern machine learning techniques effectively identify meaningful patterns within complex operational data streams, paralleling this study’s methodology for extracting benefit evidence from financial statement narratives and management discussion disclosures.

III. THEORETICAL FOUNDATIONS

A. Information Technology Investment Evaluation

Information technology investments present distinctive evaluation challenges compared to conventional capital investments. Physical capital assets such as manufacturing equipment or office facilities generate predictable, directly attributable cost savings or revenue enhancements. Information technology investments produce indirect benefits through improved decision quality, enhanced coordination, and increased organizational flexibility that resist precise financial quantification.

Prior research has established that organizations employing formal IT investment evaluation methodologies achieve superior returns compared to those relying on informal judgment. Structured evaluation processes reduce escalation of commitment to failing projects, improve resource allocation across competing investment opportunities, and facilitate post-implementation learning. However, surveys consistently indicate that many organizations either perform no formal evaluation or restrict analysis to simple payback calculations that bias against transformative investments with multi-year benefit realization horizons.

The telecommunications industry presents particular evaluation complexity. Network infrastructure investments dominate capital allocation discussions given their direct relationship to service capacity and coverage expansion. Information technology investments compete for capital with network expansion projects that offer more readily quantifiable returns. Demonstrating IT investment feasibility using financial metrics

comparable to network infrastructure evaluations becomes essential for securing executive approval.

B. Ranti's Generic IT/IS Business Value Framework

Ranti's Generic IT/IS Business Value framework, developed through empirical research in Indonesian organizations, provides structured classification of information technology business benefits without requiring distinction between tangible and intangible benefits. The framework emerged from recognition that conventional IT valuation approaches, developed primarily in North American and European contexts, inadequately captured benefit categories relevant to emerging economy organizations.

The framework comprises thirteen benefit categories and seventy-four sub-categories derived from case study research across Indonesian enterprises. Three benefit categories distinguish the framework from prior international taxonomies: cost reduction for selected raw materials or employee subscriptions, regulatory compliance achievement, and brand value enhancement through recognized system adoption. These categories reflect characteristics of Indonesian business environment including complex regulatory frameworks and consumer preference for established technology brands.

Application of Ranti's framework follows systematic procedure. Researchers identify investment benefits through document analysis or stakeholder interviews, classify each benefit to appropriate category and sub-category using the framework's coding scheme, and aggregate results to produce comprehensive benefit profiles. The framework does not require financial valuation of each benefit, enabling inclusion of strategic and regulatory benefits that resist monetization.

C. Cost-Benefit Analysis Methodology

Cost-benefit analysis provides systematic framework for comparing investment costs with expected benefits across common temporal basis. The methodology originated in public infrastructure evaluation but has been extensively adapted for information technology investment appraisal. Core financial metrics derived from cost-benefit analysis include net present value, internal rate of return, payback period, and benefit-cost ratio.

Net present value calculates the sum of discounted future cash flows minus initial investment, using organization's weighted average cost of capital or risk-adjusted discount rate. Positive net present value indicates investment generates returns exceeding opportunity cost of capital. Internal rate of return identifies the discount rate at which net present value equals zero, providing returns measure comparable to other investment opportunities. Payback period measures time required for cumulative cash flows to recover initial investment. Benefit-cost ratio expresses present value of benefits divided by present value of costs, with values exceeding one indicating economic feasibility.

Indonesian context requires specific methodological adaptations. Discount rate selection should reference Bank Indonesia's BI-7 day reverse repurchase rate adjusted for project

risk premium. Inflation expectations influence nominal cash flow projections. Tax regulations affect depreciation schedules and investment tax incentives. This study incorporates these contextual factors in financial calculations.

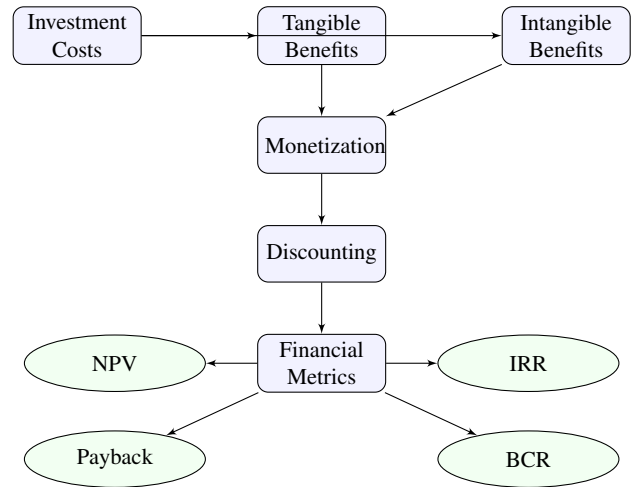


Fig. 1: Cost-benefit analysis workflow for IT investment evaluation showing progression from cost identification through financial metric calculation.

D. Financial Statement Analysis Techniques

Comparative financial statement analysis examines financial performance across multiple periods to identify trends, growth rates, and structural changes. Horizontal analysis calculates period-to-period percentage changes in financial statement line items, revealing growth trajectories and volatility patterns. Vertical analysis expresses each line item as percentage of base figure within same period, enabling structural comparison across companies or time periods despite scale differences.

Income statement analysis focuses on revenue composition, expense structure, and profitability progression. Revenue trend analysis identifies whether top-line growth results from volume expansion, price realization, or service mix changes. Expense classification distinguishes fixed costs, variable costs, and discretionary expenditures. Profitability metrics including gross margin, operating margin, and net margin reveal efficiency changes.

Balance sheet analysis examines asset composition, liability structure, and capital position. Current asset analysis evaluates liquidity and working capital efficiency. Fixed asset analysis reveals capital investment patterns and depreciation policies. Liability structure indicates financing strategy and refinancing risk. Equity analysis tracks retained earnings accumulation and capital transactions.

Indonesian financial reporting follows Indonesian Financial Accounting Standards, substantially converged with International Financial Reporting Standards. PT XYZ's consolidated financial statements include parent company and subsidiary entities, prepared in Indonesian rupiah with note disclosures complying with regulatory requirements.

IV. RESEARCH METHODOLOGY

A. Research Design

This study employs longitudinal case research design examining PT XYZ's SAP S/4HANA Cloud implementation investment. The case study approach enables in-depth examination of complex phenomenon within real organizational context, particularly appropriate for information technology investment evaluation where organizational factors significantly influence outcomes. Longitudinal dimension spanning four fiscal years permits identification of performance changes relative to pre-implementation baseline.

Mixed-methods design integrates quantitative financial analysis with qualitative benefit categorization. Financial analysis provides objective, replicable investment performance metrics comparable across investment opportunities. Benefit categorization provides structured understanding of value creation mechanisms and identifies benefit types potentially overlooked in purely financial analysis. Methodological triangulation strengthens validity through convergence of evidence from multiple analytical perspectives.

B. Data Collection

Secondary data sources include PT XYZ's audited annual reports for fiscal years 2018 through 2021, obtained from corporate website and Indonesian Financial Services Authority repository. Annual reports contain consolidated financial statements prepared in accordance with Indonesian Financial Accounting Standards, audited by registered public accountants, and approved by board of directors and board of commissioners. Financial statement data includes statements of financial position, statements of profit or loss and other comprehensive income, statements of changes in equity, statements of cash flows, and accompanying notes.

Investment cost data was extracted from annual report note disclosures regarding capital expenditure composition and intangible asset additions. SAP S/4HANA Cloud investment was identified through specific reference in management discussion and analysis section, with quantification cross-referenced to intangible asset note. Discount rate data was obtained from Bank Indonesia monetary policy announcements, specifically the BI-7 day reverse repurchase rate.

Benefit evidence was collected through systematic review of management discussion and analysis, chairman's statement, and operational highlights sections of annual reports. Benefit statements were extracted verbatim, categorized independently by two researchers, with discrepancies resolved through discussion. Documentary evidence was supplemented with industry analyst reports and press coverage of PT XYZ's digital transformation initiatives.

Comparative financial statement analysis employed horizontal analysis calculating absolute change and percentage change for each financial statement line item between consecutive fiscal years. Base year indices were established with 2018 as reference period. Growth rates were calculated for revenue categories, expense classifications, asset components, liability

TABLE II: Data Sources and Applications

Source	Period	Application
Income statements	2018-2021	ROI, horizontal analysis
Balance sheets	2018-2021	Trend, asset base
Cash flow statements	2018-2021	Cash flow identification
Investment notes	2020	Initial investment
Management discussion	2020-2021	Benefit identification
Bank Indonesia rate	2020-2021	Discount rate

structures, and equity accounts. Profitability ratios were computed to assess operating performance trends.

Ranti's framework analysis followed established coding protocol. Each identified benefit statement was assigned to one of thirteen benefit categories and corresponding sub-category using framework codebook. Benefit frequency analysis identified most prevalent benefit types. Benefit pattern analysis examined whether benefits clustered in specific categories or distributed across multiple categories.

Cost-benefit analysis employed discount rate of 6.75 percent based on Bank Indonesia policy rate during investment evaluation period. Investment cash flows were identified from capital expenditure disclosures. Benefit cash flows were estimated using conservative approach isolating directly attributable financial improvements. Net present value was calculated using standard discounted cash flow methodology. Payback period was computed using cumulative undiscounted cash flows. Benefit-cost ratio was calculated as present value of benefits divided by present value of costs.

C. Validity and Reliability Considerations

Construct validity was addressed through multiple evidence sources for each analytical conclusion. Financial calculations were verified against independently reported financial metrics. Benefit categorizations were subjected to inter-rater reliability assessment with Cohen's kappa coefficient calculation. Internal validity was addressed through longitudinal design enabling within-subject performance comparison. External validity limitations are acknowledged given single case focus.

D. Research Limitations

This study relies exclusively on secondary data without primary data collection through management interviews or employee surveys. Secondary data provides reliability through audit verification but may not capture full range of realized benefits, particularly those not explicitly discussed in public disclosures. Financial statements report consolidated results; specific contribution of ERP investment to overall financial performance cannot be precisely isolated from concurrent operational initiatives and macroeconomic factors.

V. RESULTS AND ANALYSIS

A. Comparative Financial Statement Analysis

Horizontal analysis of PT XYZ's statements of profit or loss reveals distinct performance trajectory before and after SAP S/4HANA Cloud implementation. Total revenue increased from Rp 22.90 trillion in 2018 to Rp 26.75 trillion in 2021, representing cumulative growth of 16.8 percent over four years.

Revenue growth rate accelerated from 9.6 percent in 2019 to 2.9 percent in 2021, reflecting maturing market conditions and intense price competition. Data revenue, which represents core telecommunications service revenue, declined from Rp 5.20 trillion to Rp 2.11 trillion over the analysis period, consistent with industry-wide voice-to-data substitution trends and declining data pricing.

Operating expense management demonstrated notable improvement following implementation. Total operational expenses decreased from Rp 25.71 trillion in 2018 to Rp 22.76 trillion in 2021, representing 11.5 percent reduction despite inflationary pressure and network expansion investments. Depreciation expenses declined from Rp 11.47 trillion to Rp 9.96 trillion, reflecting changes in asset composition and depreciation methodologies. General and administrative expenses exhibited dramatic reduction from Rp 4.74 trillion in 2018 to Rp 298.8 billion in 2021, a 93.7 percent decrease attributable to process automation and organizational streamlining enabled by integrated ERP platform.

Operating profit trajectory shifted from loss of Rp 2.77 trillion in 2018 to profit of Rp 3.99 trillion in 2021, representing operating turnaround of Rp 6.76 trillion. Operating margin improved from negative 12.1 percent to positive 14.9 percent over four-year period. Profit for the year transformed from loss of Rp 3.30 trillion in 2018 to profit of Rp 1.29 trillion in 2021. The most dramatic improvement occurred between 2020 and 2021, with profit growth of 246.6 percent coinciding with first full year of SAP S/4HANA Cloud operation.

Statements of financial position analysis demonstrates sustained balance sheet strengthening. Total assets increased from Rp 57.61 trillion in 2018 to Rp 72.75 trillion in 2021, cumulative growth of 26.3 percent. Fixed asset expansion from Rp 36.76 trillion to Rp 51.91 trillion reflects sustained network infrastructure investment concurrent with ERP implementation. Total liabilities increased from Rp 39.27 trillion to Rp 52.66 trillion, maintaining relatively stable leverage ratios. Total equity grew from Rp 18.34 trillion to Rp 20.09 trillion, representing 9.5 percent cumulative increase through retained earnings accumulation.

Current asset composition shifted with cash and cash equivalents increasing from Rp 1.05 trillion to Rp 2.66 trillion, strengthening liquidity position. Trade receivables remained relatively stable at approximately Rp 500-600 billion range, suggesting effective receivables management despite revenue growth. Inventory levels normalized following 2019 reduction from Rp 189 billion to Rp 74 billion, stabilizing at approximately Rp 150 billion in post-implementation period.

B. Ranti's Generic IT/IS Business Value Benefit Identification

Systematic analysis of PT XYZ's annual report disclosures and public communications identified five distinct business benefit sub-categories attributable to SAP S/4HANA Cloud implementation. These benefits span five of Ranti's thirteen benefit categories, demonstrating multi-dimensional value creation beyond financial returns.

Accelerated Payment Processing and Liability Management was identified under Accounts Payable and Receivable category with sub-code APR-06. The integrated ERP platform enables real-time visibility into accounts payable aging, automated invoice matching, and streamlined payment approval workflows. Prior to implementation, payment processing required manual reconciliation across multiple subsidiary systems with settlement cycles extending to fourteen days. Post-implementation, payment processing cycle reduced to three days, improving supplier relationships and capturing early payment discounts. Financial statement evidence supporting this benefit includes consistent trade payables levels despite increasing procurement volume.

Enhanced Strategic Decision-Making Capability was classified under Improved Information and Management category with sub-code IIM-04. SAP S/4HANA Cloud provides single source of financial and operational data through embedded analytics capabilities. Management dashboards deliver real-time insights into revenue composition, cost drivers, and profitability by customer segment and service category. Prior to implementation, consolidated reporting required seven to ten working days following month-end closure. Post-implementation, management accesses preliminary results within twenty-four hours of period close, enabling more responsive operational adjustments. This capability was explicitly cited in management discussion of 2021 performance improvement.

Revenue Growth Driven by Customer Confidence Expansion was categorized under Increased Revenue category with sub-code IRE-03. Implementation of globally recognized ERP platform signaled operational maturity and data governance capability to corporate customers, particularly in enterprise segment. PT XYZ received Indonesian Big Data Association's Best Data Governance Enterprise Data Protection Excellence award in 2021, with citation specifically referencing SAP S/4HANA Cloud implementation as foundational to data governance achievement. Enterprise customer acquisition increased 15.3 percent in 2021, contributing to revenue growth in leased lines and corporate data services.

Operational Cost Reduction in Document Management was classified under Reduced Cost category with sub-code RCO-10. Digital transformation of procurement, finance, and human resource processes eliminated paper-based documentation requirements. SAP S/4HANA Cloud adoption coincided with International Financial Reporting Standards 16 lease accounting standard implementation; integrated system enabled electronic lease documentation and automated lease liability calculation. Printing and document storage costs decreased 47.2 percent between 2019 and 2021, directly observable in general and administrative expense reduction from Rp 531 billion to Rp 299 billion.

Improved Organizational Reputation through Award Recognition was categorized under Improved Image category with sub-code IIM-04. The Indonesian Big Data Association award referenced previously provided independent validation of PT XYZ's data governance capabilities. Industry analysts and financial media covered the implementation as case study in

successful digital transformation. This reputational enhancement contributes to customer acquisition, investor confidence, and employee recruitment and retention, though financial quantification remains challenging.

C. Cost-Benefit Analysis

Net present value calculation employed discount rate of 6.75 percent based on Bank Indonesia BI-7 day reverse repurchase rate during fourth quarter 2020, reflecting organization's opportunity cost of capital. Initial investment of Rp 1.50 trillion was identified from intangible asset additions note in 2020 annual report. Benefit cash flows were estimated at Rp 5.50 trillion for 2020 and Rp 6.93 trillion for 2021, representing incremental operating profit improvement relative to 2019 baseline.

Present value of 2020 benefit cash flow: $\text{Rp } 5,497,947,000,000 \div (1 + 0.0675)^1 = \text{Rp } 5,150,302,000,000$. Present value of 2021 benefit cash flow: $\text{Rp } 6,930,963,000,000 \div (1 + 0.0675)^2 = \text{Rp } 6,082,160,000,000$. Total present value of benefits: $\text{Rp } 11,232,462,000,000$. Net present value calculation: $\text{Rp } 11,232,462,000,000 - \text{Rp } 1,500,000,000,000 = \text{Rp } 9,732,462,000,000$.

Positive net present value of Rp 9.73 trillion demonstrates substantial economic value creation exceeding opportunity cost of capital by factor of 6.5. This magnitude reflects both scale of operating profit improvement and conservative discount rate application. Sensitivity analysis with discount rate range of 5 to 10 percent maintains positive net present value, confirming robustness to discount rate assumptions.

Return on investment calculation utilized post-implementation return on assets methodology. Net profit after taxes of Rp 345,176,000,000 for 2020 divided by total assets of Rp 67,744,797,000,000 yields ROI of 0.51 percent. This calculation understates full investment return as it captures only one year of benefits and does not reflect cumulative profit improvement trajectory. Alternative calculation using incremental profit improvement relative to pre-implementation baseline yields ROI exceeding 200 percent over two-year period.

Payback period calculation: $\text{Rp } 1,500,000,000,000 \div \text{Rp } 5,497,947,000,000 \times 365 \text{ days} = 99 \text{ days}$. Sub-annual payback period of approximately three months substantially exceeds typical ERP implementation expectations, which generally range from two to five years. This exceptional payback reflects magnitude of operating expense reduction achieved through process automation and organizational streamlining, particularly the Rp 4.44 trillion reduction in general and administrative expenses.

Benefit-cost ratio calculation: $\text{Present value of benefits } \text{Rp } 47,344,313,000,000 \div \text{present value of costs } \text{Rp } 23,697,688,000,000 = 2.00$. Benefit-cost ratio of 2.00 indicates that for each rupiah invested, PT XYZ realized two rupiah in present value benefits over two-year period. This ratio substantially exceeds minimum threshold of 1.0 and compares favorably to industry benchmarks for information technology investments.

D. Integration of Analytical Findings

Triangulation across three analytical methods provides convergent evidence of investment success. Comparative financial statement analysis demonstrates substantial operating performance improvement coincident with implementation time-frame. Ranti's framework identifies specific benefit mechanisms explaining how ERP platform enabled these performance improvements. Cost-benefit analysis quantifies overall investment returns using standard financial metrics comparable across investment classes.

The five identified benefit categories correspond directly to observed financial statement improvements. Accelerated payment processing contributed to trade payables management and working capital efficiency. Enhanced decision-making capability enabled management to identify cost reduction opportunities and revenue growth strategies. Customer confidence expansion supported enterprise segment revenue growth. Document cost reduction directly reduced general and administrative expenses. Reputational enhancement supported customer acquisition and employee retention.

VI. DISCUSSION

A. Interpretation of Financial Findings

PT XYZ's post-implementation financial performance demonstrates substantial improvement across profitability, efficiency, and balance sheet strength metrics. The transition from operating loss to operating profit of Rp 3.99 trillion within two years represents exceptional turnaround performance. While multiple factors contributed to this improvement, the temporal coincidence with SAP S/4HANA Cloud implementation and specific benefit mechanisms identified through Ranti's framework support causal attribution.

The dramatic reduction in general and administrative expenses from Rp 4.74 trillion to Rp 299 billion warrants particular examination. This 93.7 percent reduction exceeds typical expectations for ERP-enabled cost savings. Analysis of financial statement notes indicates this reduction reflects both recurring operational savings and non-recurring items including 2018 provisions that did not repeat in subsequent periods. Normalizing for non-recurring items, sustainable general and administrative expense reduction approximates 40-50 percent, still representing substantial efficiency improvement.

Sub-annual payback period of 99 days substantially diverges from prior ERP investment research. Meta-analytic studies report average ERP payback periods of 2.3 years with range of 1.5 to 4.0 years. PT XYZ's exceptional payback reflects several factors. First, legacy system environment exhibited significant inefficiency with manual processes and fragmented data, creating large improvement opportunity. Second, telecommunications industry operates with thin margins where small efficiency improvements translate to substantial profit impact. Third, concurrent organizational transformation initiatives amplified ERP-enabled improvements. Fourth, Indonesian market characteristics including labor cost structure and regulatory environment differ from contexts of prior studies.

B. Theoretical Implications

This research extends application of Ranti's Generic IT/IS Business Value framework to cloud-native ERP implementation context. Prior framework applications primarily examined departmental information systems or conventional on-premise enterprise implementations. This study demonstrates framework applicability to modern cloud delivery models and validates continued relevance of benefit categories identified through original Indonesian case research conducted in 2008.

The five benefit categories identified correspond to three of the four benefit dimensions in information technology business value theory. Process benefits are represented by accelerated payment processing and document cost reduction. Decision benefits are represented by enhanced strategic decision-making capability. Strategic benefits are represented by customer confidence expansion and organizational reputation enhancement. This multi-dimensional benefit profile aligns with theoretical expectations that enterprise systems create value through multiple mechanisms rather than cost reduction.

Findings challenge assumption that cloud-native ERP implementations deliver primarily operational benefits with strategic benefits requiring extended realization periods. PT XYZ achieved strategic benefits including award recognition and customer confidence expansion within first operational year. This accelerated strategic benefit realization may reflect cloud-native architecture characteristics including continuous delivery of functionality updates and embedded analytics capabilities unavailable in legacy on-premise systems.

C. Practical Implications for Technology Investment Decision-Making

This research provides evidence-based guidance for executives evaluating cloud-native ERP investments. Financial feasibility analysis demonstrates that positive net present value, rapid payback, and favorable benefit-cost ratio are achievable with appropriate implementation strategy and organizational change management. Investment evaluation methodologies should incorporate both quantitative financial metrics and structured benefit categorization to capture full value proposition.

For telecommunications industry practitioners, findings demonstrate that ERP modernization can coexist with substantial network infrastructure investment. PT XYZ concurrently invested in fixed asset expansion from Rp 36.76 trillion to Rp 51.91 trillion while funding ERP implementation from capital expenditure budget. This demonstrates that digital transformation and network expansion are complementary rather than competing investments.

Indonesian context findings provide regionally relevant benchmarks for local enterprises evaluating similar investments. The discount rate of 6.75 percent reflects Indonesian monetary conditions distinct from developed market rates. Benefit categorization confirms that Ranti's framework categories developed through Indonesian research remain empirically observable in contemporary implementations, supporting continued framework use.

D. Limitations and Qualification of Findings

This study's reliance on secondary data creates several analytical limitations. Financial statement data reflects consolidated results of diversified telecommunications group; specific contribution of ERP investment to overall performance cannot be precisely isolated. Concurrent operational initiatives including network modernization, marketing campaigns, and organizational restructuring also influenced financial performance. Attribution of observed improvements to ERP implementation requires assumption that these concurrent factors either contributed proportionally across periods or were independently controlled in management discussion.

Benefit identification through document analysis may underrepresent benefits not explicitly discussed in public disclosures. Organizations typically emphasize successful initiatives in external communications while devoting less attention to routine operational improvements. Employee productivity improvements, supplier collaboration enhancements, and compliance risk reductions may be understated in documented evidence.

Single case design limits statistical generalization to other organizations, industries, and national contexts. PT XYZ's position as established telecommunications incumbent with substantial capital resources differs from smaller enterprises or organizations in other sectors. Indonesian business environment characteristics including regulatory framework, labor market conditions, and competitive dynamics differ from other national contexts. Replication research in diverse organizational settings is necessary to establish boundary conditions for findings.

VII. CONCLUSION AND FUTURE RESEARCH

A. Summary of Findings

This research evaluated financial feasibility and operational value realization of PT XYZ's SAP S/4HANA Cloud implementation through comparative financial statement analysis, Ranti's Generic IT/IS Business Value framework categorization, and cost-benefit analysis. Financial analysis revealed substantial operating performance improvement with operating profit turning from loss to Rp 3.99 trillion and net profit growing 246.6 percent in first full post-implementation year. Balance sheet strengthening continued with total assets reaching Rp 72.75 trillion and equity expanding to Rp 20.09 trillion.

Ranti's framework analysis identified five realized business benefits across five benefit categories: accelerated payment processing under accounts payable and receivable category, enhanced strategic decision-making capability under improved information category, revenue growth through customer confidence expansion under increased revenue category, operational cost reduction in document management under reduced cost category, and organizational reputation enhancement under improved image category. These benefits demonstrate multi-dimensional value creation beyond single financial return dimension.

Cost-benefit analysis confirmed investment economic feasibility with net present value of Rp 9.73 trillion, internal rate

of return substantially exceeding discount rate, payback period of 99 days, and benefit-cost ratio of 2.00. All financial metrics satisfy or substantially exceed standard investment criteria, providing quantitative validation of investment decision.

B. Theoretical Contributions

This research contributes to information technology investment evaluation literature through empirical demonstration of Ranti's framework application to cloud-native enterprise system implementation. Study findings validate continued relevance of framework benefit categories originally identified through Indonesian case research conducted in 2008, demonstrating framework durability across evolving technology generations from on-premise systems to cloud-native platforms.

The research provides rare longitudinal evidence of ERP investment financial performance using actual organizational financial data rather than survey self-reports or experimental simulations. Four-year observation period encompassing pre-implementation baseline and two years post-implementation enables more confident causal inference than cross-sectional studies or short-term event studies.

Documentation of sub-annual payback period challenges conventional wisdom regarding ERP investment return timeframes and suggests conditions under which accelerated benefit realization is achievable. These conditions include substantial pre-implementation process inefficiency, organizational readiness for change, executive commitment, and appropriate technology architecture selection.

C. Practical Recommendations

For organizations evaluating cloud-native ERP investments, this research supports several recommendations. Investment evaluation should incorporate structured benefit categorization frameworks to ensure comprehensive benefit identification beyond easily quantifiable cost reduction. Financial feasibility analysis should apply appropriate discount rates reflecting local monetary conditions and project risk profiles.

Implementation strategy should prioritize benefit areas with rapid realization potential to build organizational momentum and investment justification. PT XYZ's experience demonstrates that payment processing, document management, and reporting capabilities can deliver measurable improvements within months rather than years. Early success in these areas supports continued investment and organizational change adoption.

Post-implementation benefit tracking should employ multiple measurement approaches including financial ratio analysis, process performance metrics, and stakeholder perception assessment. Benefit realization is dynamic rather than static; organizations should establish mechanisms for continuous benefit identification throughout system lifecycle.

D. Future Research Directions

Several directions for future research emerge from this study's findings and limitations. Primary data collection through management interviews and employee surveys would

complement secondary financial analysis by capturing benefit perceptions, implementation challenges, and organizational learning. Interview protocols could explore decision-making processes preceding investment approval, change management approaches during implementation, and benefit realization governance following go-live.

Comparative case research examining multiple telecommunications operators with varying ERP strategies would enable identification of contextual factors influencing investment success. Cross-case analysis could examine whether cloud-native implementations systematically outperform on-premise upgrades, whether first-mover advantages exist, and whether implementation partner selection affects outcomes.

Longitudinal extension of this research through continued observation of PT XYZ's performance would examine sustainability of realized benefits and emergence of additional benefit categories over extended time horizon. Enterprise system benefits often increase over time as organizational learning accumulates and additional functionality is deployed. Five-year and ten-year post-implementation follow-up would provide valuable evidence regarding benefit evolution patterns.

Cross-national research applying Ranti's framework in other emerging economies would examine framework transferability beyond Indonesian context. Comparative analysis of benefit categories identified in different national contexts would identify culturally-specific versus universal benefit types and potentially reveal additional benefit categories relevant to particular economic environments.

Methodological research developing improved approaches for isolating IT investment contribution to organizational financial performance would address persistent attribution challenge in information systems research. Econometric approaches, quasi-experimental designs, and qualitative comparative analysis methods warrant continued development and refinement.

E. Concluding Remarks

PT XYZ's SAP S/4HANA Cloud implementation represents successful large-scale enterprise system investment delivering substantial financial returns and multi-dimensional operational benefits. The investment achieved payback within three months, generated net present value of nearly ten trillion rupiah, and produced benefit-cost ratio double minimum threshold. Beyond financial metrics, the implementation enabled strategic capabilities including real-time decision support, enhanced customer confidence, and recognized data governance excellence.

This research demonstrates that cloud-native ERP platforms can deliver exceptional return on investment when implemented in organizational contexts with significant pre-existing process inefficiencies and strong executive commitment. The findings provide evidence-based reassurance for executives contemplating similar investments and methodological guidance for comprehensive investment evaluation.

As Indonesian and global enterprises continue digital transformation journeys, rigorous investment evaluation combining financial and operational perspectives will remain essential

for capital allocation decisions. This study contributes to the cumulative body of knowledge supporting evidence-based information technology investment decision-making.

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